RAJ BHARAT RASAL

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Second-year B.Tech student focused on backend systems architecture, distributed computing, and multi-agent AI pipelines. I design and build complex infrastructure from the ground up—currently developing a distributed tensor-sharding network and a multi-agent communication system—prioritizing system design, integration logic, and scalable architecture.

EDUCATION

ITM Skills University Aug 2025 – May 2029

B.Tech in Computer Science Engineering

10th CBSE: 96% | 12th CBSE: 83.4%

EXPERIENCE

Independent Tutor Ongoing

Mathematics & Physics (Grades 10–12)

- Taught 50+ students, translating complex scientific concepts into clear, structured lessons.

PROJECTS

SATURDAY / SUNDAY — Distributed Compute Network

Architect & Developer (Private Repo, demo on request)

- Built a distributed compute network to run large local LLMs, solving hardware limitations of processing complex AI models on single devices.
- Implemented automatic hardware discovery and tensor sharding to split a single model's processing across heterogeneous Mac/Windows nodes via LAN and Tailscale VPN.
- Designed a zero-trust communication protocol between nodes, utilizing proof-of-compute verification to reject nodes returning corrupted results.

F.R.I.D.A.Y — Multi-Agent Communication Backend

Lead Backend & Architecture (Private Repo, demo on request)

- Originated at SummerHacks 2026 (team of 4); currently driving ongoing backend development.
- Designed a decoupled 3-agent architecture (Herald, Nexus, Dispatch) enforcing strict separation of concerns so agents never call each other directly.
- Implemented LLM intent classification with confidence-based routing logic to catch ambiguous queries and trigger clarification flows.
- Tech Stack: Python, FastAPI, Baileys API (WhatsApp), APScheduler

EduOS — AI Curiosity-First Education Engine

Creator (Private Repo, demo on request)

- Developed an AI platform featuring a "pinch engine" that generates one high-value conceptual knowledge connection per day to students.
- Engineered core logic prioritizing curiosity-driven content delivery before standard notes.
- Built WhatsApp API integration, deliberately decoupled during development to prevent unauthorized message broadcasting.

ACHIEVEMENTS

- 3rd Place: Deal Dome Competition — awarded for the architecture of SecureVerse.
- Security Research: Identified Open Redirect vulnerability in OPPO Find X3 Pro (HackerOne).